THE OPEN SOURCE AUDIT: AN ANALYSIS OF PYTHON

A Capstone Project for the OSS NGMC Course

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Software Audited: Python (Managed by PSF)

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1. INTRODUCTION

In the contemporary era of cloud computing, artificial intelligence, and global automation, Open Source Software (OSS) has transitioned from a niche hobbyist movement into the backbone of the global economy. This report provides a structured audit of **Python**, a high-level, interpreted programming language that epitomizes the collaborative and transparent nature of the open-source movement.

Python was selected for this audit because of its unique position as both a "glue language" for Linux systems and a primary tool for modern data science. This audit examines Python's origins, its permissive licensing model, its footprint on the Linux filesystem, and its role within the broader Free and Open Source Software (FOSS) ecosystem.

PART A — ORIGIN AND PHILOSOPHY (Units 1 & 2)

A1. The Problem This Software Was Created to Solve

To understand Python, one must understand the landscape of computing in the late 1980s. At the time, developers faced a "Grand Binary": you could either write code in low-level languages like C, which was powerful but extremely complex and time-consuming, or use Shell scripts, which were easy for automation but lacked the structures (like classes or advanced data types) required for complex applications.

Guido van Rossum, working at CWI in the Netherlands, identified a "missing middle." He wanted a language that was as readable as English but as capable as C. He drew inspiration from a language called ABC, which focused on teaching non-programmers but lacked features like file-system access or error handling.

The Solution:
Python was designed to be "extensible." If a task was too slow in Python, a developer could write that specific module in C and "plug it in." This allowed Python to remain simple at the surface while being infinitely powerful underneath. The origin of Python is thus a story of pragmatism over perfection—a tool built by a developer to solve his own frustration with existing tools.

A2. The License — The Python Software Foundation (PSF) License

Python’s license is a "Permissive" license, which stands in contrast to "Copyleft" licenses like the GPL (General Public License).

The Four Freedoms Analysis:

1. Freedom 0 (Run): Python can be run for any purpose, from academic research to high-frequency trading on Wall Street.
2. Freedom 1 (Study): The CPython source code is hosted on GitHub, allowing anyone to study the "Global Interpreter Lock" (GIL) or memory management systems.
3. Freedom 2 (Redistribute): Users can bundle Python with their own software and sell it.
4. Freedom 3 (Improve): Anyone can submit a "PEP" (Python Enhancement Proposal) to suggest changes to the language.

GPL vs. PSF License: The Critical Difference

The GPL license is "infectious"; if you use GPL code in your project, your project must also be open source. The PSF license, however, allows for Proprietary Integration. A company like Apple or Google can take Python, modify it for their internal systems, and they are *not* legally required to share those changes. This makes Python more attractive to corporations, leading to faster adoption and better funding for the Python Software Foundation.

A3. The Ethics of Open Source

The ethics of open source are rooted in the "Cathedral vs. the Bazaar" model. Proprietary software is like a Cathedral—built by a closed group of elite architects behind locked doors. Open source is a Bazaar—a messy, loud, but highly innovative marketplace where everyone contributes.

The Argument for Open Knowledge:
Is it ethical for a corporation to profit from the labor of thousands of volunteers? In the Python community, the answer is found in the **"Social Contract."** While companies profit, they also contribute back by hiring core developers to work on the language full-time. This creates a "symbiotic" relationship where commerce and community coexist.

Standing on the Shoulders of Giants:
In Python, no one starts from zero. If you build a website, you use Django. If you do AI, you use TensorFlow. This ethical commitment to "not reinventing the wheel" is what has allowed software to evolve at an exponential rate compared to other engineering fields.

PART B — LINUX FOOTPRINT (Unit 2)

The Resident Interpreter
Python is not an "external" application on Linux; it is a core component. On a system like Ubuntu or Fedora, Python is used for:
1. System Maintenance: Tools like `apt` (package manager) rely on Python scripts.
2. Automation: The Linux "Cloud-Init" process and many installer scripts (like "Anaconda") are Python-based.

Key Directories:
- Binary: `/usr/bin/python3` — The executable that reads your script.
- Standard Library: `/usr/lib/python3.10/` — Contains the "Batteries Included" modules like `os`, `sys`, and `math`.
- Third-Party Packages: `/usr/local/lib/python3.x/dist-packages/` — Where libraries installed via `pip` are stored.

Update Model:
Python uses a "Rolling Stability" model. Minor versions (3.10 to 3.11) are released annually, while security patches are released as needed. On Linux, these updates are managed by the **OS Maintainers**, ensuring that an update to Python doesn't break the entire operating system.

PART C — THE FOSS ECOSYSTEM (Units 3 & 4)

Python and the LAMP Stack

Historically, the "P" in the LAMP stack stood for Perl or PHP. Today, Python is the dominant "P." It integrates seamlessly with:

- Linux: As the host OS.
- Apache/Nginx: Via "WSGI" or "ASGI" interfaces.
- MySQL/PostgreSQL: Through robust drivers like `psycopg2`.

Community and Forks

Python’s open-source nature has led to specialized versions:

1. MicroPython: For the Internet of Things (IoT).
2. PyPy: For high-performance computing.
3. CircuitPython: For educational robotics.

The community is governed by a Steering Council, a model that ensures no single corporation (like Microsoft or Google) can control the language's future.

PART D — OPEN SOURCE VS PROPRIETARY (Critical Analysis)

Dimension	Python (Open Source)	MATLAB (Proprietary)
Security	Highly transparent; code is audited by thousands of eyes.	Security relies on the vendor's internal audits (Security by Obscurity).
Cost	Free for everyone.	Licenses cost thousands of dollars per user.
Support and Reliability	Community forums, StackOverflow, and GitHub.	Dedicated corporate support and documentation.
Freedom to modify	You own your code and the tool used to run it.	You are "renting" the right to use the software.
Verdict	Python wins on accessibility and innovation.	MATLAB wins on specialized, "out-of-the-box" engineering tools.
Community vs Corporate Control	Governed by the non-profit PSF and community.	Entirely controlled by MathWorks.

5. SHELL SCRIPT DOCUMENTATION

Script 1: System Identity Report

Concepts Used: Command substitution, Variables, `/etc/os-release` parsing.
Logic: This script uses the `uname` and `uptime` commands to provide a snapshot of the environment where Python is running.

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Swastik | Course: Open Source Software

# --- Variables ---
STUDENT_NAME="Swastik"
SOFTWARE_CHOICE="Python"

# --- System info ---
KERNEL=$(uname -r)
USER_NAME=$(whoami)
USER_HOME=$HOME
UPTIME=$(uptime -p)
DISTRO=$(grep "HANDSOME_NAME" /etc/os-release | cut -d'=' -f2 | tr -d '"')
DATE_TIME=$(date)

```

Script 2: FOSS Package Inspector

Concepts Used: `if-then-else`, `case` statements, `dpkg` integration.
Logic: This script demonstrates "conditional logic" by checking the system's package database to see if Python is installed.

```
root@SHABI:/mnt/c/Users/shabi/OneDrive/Desktop/OSSproject# ./script2.sh
FOSS Package: python3 is installed.
Version: 3.12.3-0ubuntu2.1 | Summary: amd64 interactive
----------------------------------------
Software: Python 3
Purpose : A versatile high-level programming language.
Mission : Designed for readability and ease of use.
Open Source: Managed by the Python Software Foundation (PSF).
----------------------------------------
```

Script 3: Disk and Permission Auditor

Concepts Used: `for` loops, Array iteration, `awk` for string parsing.
Logic: This script iterates through an array of critical system paths to audit security permissions and disk usage.

```
root@SHABI:/mnt/c/Users/shabi/OneDrive/Desktop/OSSproject# ./script3.sh
==========================================
      Directory Audit Report
==========================================
/etc => Permissions: drwxr-xr-x | Owner: root | Size: 6.4M
/var/log => Permissions: drwxrwxr-x | Owner: root | Size: 435M
/home => Permissions: drwxr-xr-x | Owner: root | Size: 5.1G
/usr/bin => Permissions: drwxr-xr-x | Owner: root | Size: 210M
/tmp => Permissions: drwxrwxrwt | Owner: root | Size: 56K
------------------------------------------
Searching for the Python software location...
Python is located at: /home/linuxbrew/.linuxbrew/bin/python3
lrwxrwxrwx 1 myuser myuser 40 Dec 30 10:33 /home/linuxbrew/.linuxbrew/bin/python3 -
> ../Cellar/python@3.14/3.14.2/bin/python3
==========================================
```

Script 4: Log File Analyzer

Concepts Used: `while read` loops, Positional parameters (`$1`, `$2`), `grep`.
Logic: This script processes large text files line-by-line, demonstrating the efficiency of shell scripting for diagnostic tasks.

```
root@SHABI:/mnt/c/Users/shabi/OneDrive/Desktop/OSSproject# ./script4.sh
==========================================
 Analyzing Log File: /var/log/syslog
 Searching for Keyword: error
==========================================
------------------------------------------
Keyword 'error' found 18 times in /var/log/syslog.
Last 5 matches found:
2026-03-28T11:48:53.079964+00:00 SHABI systemd[1]: apport-autoreport.timer - Proces
s error reports when automatic reporting is enabled (timer based) was skipped becau
se of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2026-03-28T11:48:53.080727+00:00 SHABI kernel: RAS: Correctable Errors collector in
itialized.
2026-03-28T11:49:25.856654+00:00 SHABI kernel: WSL (246) ERROR: CheckConnection: ge
taddrinfo() failed: -5
2026-03-28T11:52:17.361419+00:00 SHABI systemd[1]: apport-autoreport.path - Process
 error reports when automatic reporting is enabled (file watch) was skipped because
 of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2026-03-28T11:52:17.361426+00:00 SHABI systemd[1]: apport-autoreport.timer - Proces
s error reports when automatic reporting is enabled (timer based) was skipped becau
se of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
==========================================
```

Script 5: Manifesto Generator

Concepts Used: User input (`read`), String concatenation, File redirection (`>>`).
Logic: An interactive script that creates a persistent record of the user's philosophy, demonstrating how automation can interact with human users.

```
root@SHABI:/mnt/c/Users/shabi/OneDrive/Desktop/OSSproject# ./script5.sh
==========================================
      Welcome to the Manifesto Generator
==========================================
Answer three questions to create your open source philosophy.

1. Which open source tool is most important to you? (e.g., Python, Linux, VSCode):
VSCode
2. What is the most important freedom in open source? (e.g., Modify, Share, Study):
 Modify
3. Why do we need open source software in the future? (one sentence): It promotes i
nnovation, security, affordability and global collaboration by allowing anyone to u
se,improve and share technology freely

Manifesto saved to: manifesto_root.txt
Content of your manifesto:
------------------------------------------
      Open Source Manifesto - 28 March 2026
------------------------------------------
I, root, believe that VSCode is essential for progress.
The freedom to Modify allows software to evolve beyond corporate limits.
In the future, open source matters because It promotes innovation, security, afford
ability and global collaboration by allowing anyone to use,improve and share techno
logy freely.
------------------------------------------
Generated by Shabi's OSS Script.
==========================================
root@SHABI:/mnt/c/Users/shabi/OneDrive/Desktop/OSSproject#
```

CONCLUSION

This audit has demonstrated that Python is not just a language, but a testament to the power of open-source ethics. From its origin as a developer's hobby to its current status as a global standard, Python remains true to its core values: openness, simplicity, and community.